

Output:

Name: Ravi

Balance: 25000.0

Experiment 11: Understanding static

Program 11.1: Static Variable

```
class Student {  
    String name;  
    static int count = 0;  
    Student(String name) {  
        this.name = name;  
        count++;  
    }  
    void display() {  
        System.out.println("Name: " + name);  
    }  
}  
  
public class StaticVariable {  
    public static void main(String[] args) {  
        Student s1 = new Student("Raina");  
        Student s2 = new Student("Rinku");  
        Student s3 = new Student("Amar");  
        s1.display();  
        s2.display();  
        s3.display();  
        System.out.println("Total Students: " + Student.count);  
    }  
}
```

Output:

Name: Raina

Name: Rinku

Name: Amar

Total Students: 3

Program 11.2: Static Methods

```
class Utility {  
    static int square(int n) {  
        return n * n;  
    }  
    static int cube(int n) {  
        return n * n * n;  
    }  
    static long factorial(int n) {  
        long fact = 1;  
        for (int i = 1; i <= n; i++) {  
            fact *= i;  
        }  
        return fact;  
    }  
}  
  
public class StaticMethods {  
    public static void main(String[] args) {  
        int n = 5;  
        System.out.println("Square: " + Utility.square(n));  
        System.out.println("Cube: " + Utility.cube(n));  
        System.out.println("Factorial: " + Utility.factorial(n));  
    }  
}
```

Output:

Square: 25

Cube: 125

Factorial: 120

Experiment 12: Introducing final

Program 12.1: Final Keyword

```
class Parent {  
    final int VALUE = 100;  
    final void display() {  
        System.out.println("This is a final method.");  
        System.out.println("Final variable: " + VALUE);  
    }  
}  
  
final class FinalClass {  
    void show() {  
        System.out.println("This is a final class.");  
    }  
}  
  
public class FinalDemo {  
    public static void main(String[] args) {  
        Parent p = new Parent();  
        p.display();  
        FinalClass f = new FinalClass();  
        f.show();  
    }  
}
```

Output:

This is a final method.

Final variable: 100

This is a final class.

Program 12.2: Blank Final Variable

```
class Employee {  
    final int employeeId;  
    String name;  
    Employee(int employeeId, String name) {  
        this.employeeId = employeeId;  
        this.name = name;  
    }  
    void display() {  
        System.out.println("Employee ID: " + employeeId);  
        System.out.println("Name: " + name);  
    }  
}  
  
public class BlankFinal {  
    public static void main(String[] args) {  
        Employee e1 = new Employee(101, "Raina");  
        Employee e2 = new Employee(102, "Rinku");  
        e1.display();  
        System.out.println();  
        e2.display();  
    }  
}
```

Output:

Employee ID: 101

Name: Raina

Employee ID: 102

Name: Rinku

Experiment 13:

Program 13.1: College and Department

```
class College {  
    static class Department {  
        String name;  
        Department(String name) {  
            this.name = name;  
        }  
        void display() {  
            System.out.println("Department: " + name);  
        }  
    }  
}  
  
public class CollegeDemo {  
    public static void main(String[] args) {  
        College.Department d =  
            new College.Department("CSE-AIML");  
        d.display();  
    }  
}
```

Output:

Department: CSE-AIML

Program 13.2: Employee Address

```
class Employee {  
    int id;
```

```

String name;
Employee(int id, String name) {
    this.id = id;
    this.name = name;
}
static class Address {
    String city;
    String state;
    Address(String city, String state) {
        this.city = city;
        this.state = state;
    }
    void display() {
        System.out.println("City: " + city);
        System.out.println("State: " + state);
    }
}
}
public class EmployeeAddress {
    public static void main(String[] args) {
        Employee e = new Employee(101, "Ravi");
        Employee.Address address =
            new Employee.Address("Vijayawada", "Andhra Pradesh");
        System.out.println("Employee ID: " + e.id);
        System.out.println("Name: " + e.name);
        address.display();
    }
}

```

Output:

Employee ID: 101

Name: Ravi

City: Vijayawada

State: Andhra Pradesh

Experiment 14: Inner Classes

Program 14.1: Student Address

```
class Student {  
    String name;  
    int rollNo;  
    Student(String name, int rollNo) {  
        this.name = name;  
        this.rollNo = rollNo;  
    }  
    class Address {  
        String city;  
        String state;  
        Address(String city, String state) {  
            this.city = city;  
            this.state = state;  
        }  
        void display() {  
            System.out.println("Student Name: " + name);  
            System.out.println("Roll Number: " + rollNo);  
            System.out.println("City: " + city);  
            System.out.println("State: " + state);  
        }  
    }  
}
```

```
public class StudentAddress {  
    public static void main(String[] args) {  
        Student s = new Student("Ravi", 101);  
        Student.Address a =  
            s.new Address("Vijayawada", "Andhra Pradesh");  
        a.display();  
    }  
}
```

Output:

Student Name: Ravi

Roll Number: 101

City: Vijayawada

State: Andhra Pradesh

Program 14.2: Library Management

```
class Library {  
    String libraryName;  
    Library(String libraryName) {  
        this.libraryName = libraryName;  
    }  
    class Book {  
        String title;  
        String author;  
        Book(String title, String author) {  
            this.title = title;  
            this.author = author;  
        }  
        void display() {  
            System.out.println("Library: " + libraryName);  
            System.out.println("Book Title: " + title);  
        }  
    }  
}
```

```
        System.out.println("Author: " + author);
    }
}

public class LibraryManagement {
    public static void main(String[] args) {
        Library library = new Library("Central Library");
        Library.Book book =
            library.new Book("Java Programming", "James Gosling");
        book.display();
    }
}
```

Output:

Library: Central Library

Book Title: Java Programming

Author: James Gosling